

Course: B. Tech. Branch : Mechanical Engineering Semester : III

Subject Code & Name: (BTMC303) Thermodynamics

Max Marks: 60

Date: 06-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Attempt any two sub questions from each question
3. Use of non-programmable scientific calculators is allowed.
4. Illustrate your answer with neat sketches, diagram, etc., wherever necessary.
5. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

A) Clearly differentiate between the following with suitable examples

Remembering

6

- i) Control Mass and Control Volume
- ii) Microscopic and Macroscopic
- iii) Path function and point function

B) i) A barometer measures 760 mm Hg at street level and 735 mm Hg on top of a building. How tall is the building if we assuming air density of 1.15 kg/m³.

Analyzing

6

ii) The limiting value of the ratio of the pressure of gas at the steam point and at the triple point of water when the gas is kept at constant volume is found to be 1.36605. What is the ideal gas temperature of steam point.

C) i) State the zeroth law of thermodynamics and define thermometric property.

Understanding

6

ii) Obtain the expression for quasi static pdv work for isothermal process.

Q.2 Solve Any Two of the following.

12

A) State first Law of thermodynamics for a cyclic process and show that it is consistent with the following situation.

Remembering

6

Process	Q (kJ)	W (kJ)
1-2	443	255
2-3	- 412	-340
3-4	- 222	0
4-1	255	149

B) Started from first principle show that throttling is an isenthalpic process by apply SFEE.

Understanding

6

C) Air flows steadily at the rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and a specific volume of 0.85 m³/kg and

Analyzing

6

leaving at 4.5 m/s with a pressure of 6.9 bar and a specific volume of 0.16 m³/kg. The internal energy of the air leaving is 88 kJ/kg greater than that of the air entering. Cooling water jackets surrounding the cylinder absorbs heat from the air at the rate of 59 W.

Q. 3 Solve Any Two of the following.		12
A) Define Heat Pump and refrigerator with suitable diagram and show that $COP_{HP} = COP_{Ref} + 1$	Understanding	6
B) State the entropy principal and explain the entropy balance and with proper illustration and define the concept of entropy generation.	Understanding	6
C) A heat pump is to be used to heat a house in winter and then reversed to cool the house in summer. The interior temperature is to be maintained at 20 °C. Heat transfer through the walls and roof is estimated to be 0.525 kJ/s per degree temperature difference the inside and outside. If the outside temperature in winter is 5 °C, what is the minimum power required to drive the heat pump.	Analyzing	6
Q.4 Solve Any Two of the following.		12
A) Define ideal gas. What is Universal gas constant? Obtain its value.	Remembering	6
B) A mass of 1 kg of air initially at 5 bar and 100 °C undergoes a cycle consisting of constant pressure expansion until the volume is doubled followed by constant volume cooling followed by the reversible adiabatic compression to the initial state. Find the net work done in the cycle.	Analyzing	6
C) Show that for an ideal gas $c_p - c_v = R$.	Analyzing	6
Q. 5 Solve Any Two of the following.		12
A) Explain the working of the throttling calorimeter with the schematic diagram and process representation on h-s diagram.	Remembering	6
B) Steam initially at 1.5 MPa, 300 °C expands reversibly and adiabatically in a steam turbine to 40 °C. Determine the ideal work output of the turbine per kg of steam.	Analyzing	6
C) Define the following; Dryness Fraction, saturated states, critical states	Remembering	6

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